General geometric 3-torsion on actual curves

Lean proves exactly nine actual points P with 3P=0 on every curve $y^2=x^3+ax+b$ over an algebraically closed field with $2\neq0$, $3\neq0$ and $4a^3+27b^2\neq0$. This includes the identity. No rank, basis, Hasse or pairing premise is assumed. Whole gate: 20 declarations, zero failures, zero trusted-native exceptions. [pcc:unverified]

The quartic is $\psi_3 = 3X^4 + 6aX^2 + 12bX - a^2$, with derivative $12(X^3 + aX + b)$. The identity

$$\psi_3(x) = 12x(x^3 + ax + b) - (3x^2 + a)^2$$

shows that a common cubic/quartic root would force $a=-3x^2$, $b=2x^3$ and discriminant zero. Hence the quartic is separable, of degree four, and has four distinct roots. The cubic value is nonzero at each root; each $y^2=x^3+ax+b$ fibre has two distinct roots. [pcc:unverified]

Actual proof step	Declaration in ThreeTorsionRankAudit
No common cubic/quartic root	quartic_cubic_no_common_root
Separable quartic and four roots	quartic_separable; quartic_root_card
Actual group torsion iff quartic zero	affine_three_torsion_iff
Construct actual nonsingular root-pair point	root_pair_nonsingular
Explicit inverse affine coordinate maps	affineRootEquiv
Include the actual group identity	geometricCoordinateEquiv
Cardinality nine	geometric_three_torsion_cardinality

The repeated-S_3 equivalence is inherited from checked actual Mathlib point arithmetic. If 3P=0, then 2P=−P has the same x-coordinate. Conversely the same-x alternatives are 2P=P or 2P=−P; the first forces an affine point to be the identity and is impossible. The coordinate equivalence precedes the count: actual torsion equals identity plus four two-point fibres. [pcc:unverified]

Source: Balasubramanian–Koblitz1998, Theorem1/proof/Remark1, printed142–143, reread. PDF SHA256 ac837b8edc47b346dbb341f787c41de0402c95371a159f9786b747bb2c987499. Its matrix argument uses geometric torsion rank before the determinant. This check proves the r=3 cardinality component under explicit short-curve characteristic conditions. The qualified source is not refuted. All-prime rank, determinant, generic BK, p<500 census and canonical pairing/divisor identification remain separate. [pcc:unverified]

Lean4.35.0 / Mathlib56a00585, only propext, Classical.choice and Quot.sound. No holes, custom axioms, native_decide, waivers or raised elaboration limits. All failed sources/logs/exits and the complete gate are retained with byte-for-byte archive checks. KB unavailable (MCP absent/wrapper127); bounded original-paper/repository/Mathlib fallback. Exact source and packet receipts accompany this report. Independent signed standing remains unverified. No native experiment or paid compute. [pcc:unverified]